Unit 4, Lesson 5: Evaluating Numerical Expressions with Integer Exponents – Part 2

Benchmark

MA.8.NSO.1.3 Extend previous understanding of the Laws of Exponents to include integer exponents. Apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to integer exponents and rational number bases, with procedural fluency.

Learning Target

- ☐ **I can evaluate a numerical expression using the laws of exponents.**

4.5.1 Warm-Up: Mistakes Help You Grow

1. Two artists are drawing a plant. As they go through the process, both artists make mistakes and start over on their drawings. One of the artists crumples their papers up and throws them on the floor. The other artist hangs them on the wall next to where she is working.

What do you think is the purpose of hanging up the drawings with mistakes?

4.5.2 Guided Practice: Evaluating Exponential Expressions

1. Danny and Reina evaluated the expression $\frac{\left(\frac{3}{4}\right)^{-3}\cdot\left(\frac{3}{4}\right)^{0}}{\left(\frac{3}{4}\right)^{-2}}$. They found the same value but used different processes. Their work is shown.

Danny's Work	Reina's Work
$\frac{\left(\frac{3}{4}\right)^{-3}\cdot\left(\frac{3}{4}\right)^{0}}{\left(\frac{3}{4}\right)^{-2}}$	$\frac{\left(\frac{3}{4}\right)^{-3}\cdot\left(\frac{3}{4}\right)^{0}}{\left(\frac{3}{4}\right)^{-2}}$
$=\frac{\left(\frac{3}{4}\right)^{-3+0}}{\left(\frac{3}{4}\right)^{-2}}$	$=\frac{\left(\frac{3}{4}\right)^{-3}\cdot 1}{\left(\frac{3}{4}\right)^{-2}}$
$=\frac{\left(\frac{3}{4}\right)^{-3}}{\left(\frac{3}{4}\right)^{-2}}$	$=\frac{\left(\frac{3}{4}\right)^{-3}}{\left(\frac{3}{4}\right)^{-2}}$
$=\left(\frac{3}{4}\right)^{-3-(-2)}$	$=\left(\frac{3}{4}\right)^{-3-(-2)}$
$=\left(\frac{3}{4}\right)^{-1}$	$=\left(\frac{3}{4}\right)^{-1}$
$=\frac{4}{3}$	$=\frac{4}{3}$

a. With your partner, discuss and compare how each student evaluated the expression.

b. **Use words or phrases from the bank to complete the descriptions of each student's work. Some words or phrases will be used more than once and others not at all.**

zero	identity
power of a power	negative
power of a product	product of powers
power of a quotient	quotient of powers

Danny applied the ______________________________ exponent law first by rewriting $\left(\frac{3}{4}\right)^{-3} \cdot \left(\frac{3}{4}\right)^{0}$ as $\left(\frac{3}{4}\right)^{-3}$. Then, he applied the ____________________________ exponent law, subtracting the exponents -3 and -2. Finally, he applied the _______________________ exponent law to find the value of the expression.

Reina applied the ______________________________ exponent law first by re-writing $\left(\frac{3}{4}\right)^{0}$ as 1. Then, she multiplied $\left(\frac{3}{4}\right)^{-3}$ by 1. Next, she applied the ______________________________ exponent law by subtracting the exponents -3 and -2. Finally, she applied the _______________________ exponent law to find the value of the expression.

While the students applied the laws differently, they both generated equivalent expressions in each step to find the value of the expression.

4.5.3 Try It: Evaluating Expressions

1. **Consider the expression** [illegible].

   a. **Evaluate the expression in two different ways using the strategies described in the table. Show your work under each strategy.**

Strategy 1: Start by applying the product of powers law of exponents.	**Strategy 2:** Start by applying the power of a product law of exponents.
[illegible]	[illegible]

   b. **Explain whether or not the quotient of powers law of exponents could have been applied first.**

4.5.4 Collaboration: Choose Your Own Adventure!

1. Consider the expression $\frac{(z^{-1})^4 \cdot \left(\frac{1}{z}\right)^{-4}}{z \cdot z^4}$.

a. Discuss with your partner which exponent laws could be applied in the first step. Check off all that apply from the list.

- ☐ zero exponent
- ☐ identity exponent
- ☐ negative exponent
- ☐ power of a power
- ☐ product of powers
- ☐ power of a product
- ☐ quotient of powers
- ☐ power of a quotient

b. Evaluate the expression beginning with a different first step than your partner.

c. Compare your work with your partner's and come to a consensus about the value of the expression. Discuss the similarities and differences in your work.

2. Consider the expression $\dfrac{\left(\frac{?}{?}\right)^{-4}\cdot\left(\left(\frac{?}{?}\right)^{7}\right)^{0}}{\left(\left(\frac{?}{?}\right)^{2}\right)^{3}}$.

a. Discuss with your partner which exponent laws could be applied in the first step. Check off all that apply from the list.

- ☐ zero exponent
- ☐ identity exponent
- ☐ negative exponent
- ☐ power of a power
- ☐ product of powers
- ☐ power of a product
- ☐ quotient of powers
- ☐ power of a quotient

b. Evaluate the expression beginning with a different first step than your partner.

c. Compare your work with your partner's and come to a consensus about the value of the expression. Discuss the similarities and differences in your work.

4.5.5 Your Turn!

1. Otis is evaluating the expression $\frac{12^{-4}\cdot 12^{0}}{(12^{-3})^{2}}$.

   a. Circle the expressions that show a correct first step in evaluating the expression.

   $\frac{12^{-4}\cdot 1}{(12^{-3})^{2}}$ $\qquad$ $\frac{12^{-8}\cdot 12^{0}}{12^{-6}}$ $\qquad$ $\frac{12^{-4}}{(12^{-3})^{2}}$ $\qquad$ $\frac{12^{-4}\cdot 12^{0}}{12^{-6}}$

   b. Evaluate $\frac{12^{-4}\cdot 12^{0}}{(12^{-3})^{2}}$.

2. Select all the expressions that are equivalent to $\frac{1}{25}$.

   ☐ $(5^{3})\cdot(5^{-1})^{5}$

   ☐ $\frac{(5^{[illegible]})(5^{[illegible]})}{(5^{-[illegible]})(5^{-[illegible]})}$

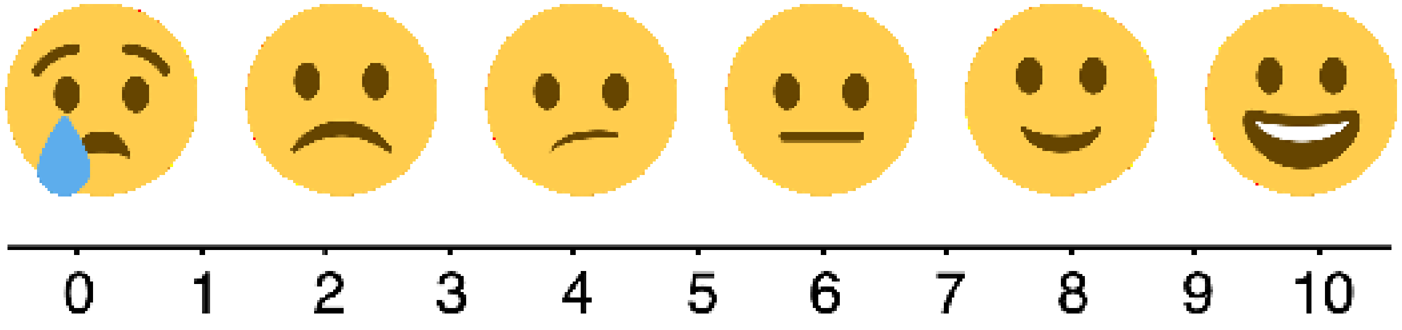
   ☐ $\frac{(5^{0})^{4}\cdot(5^{-1})}{5}$

   ☐ $\frac{(5^{[illegible]})(5^{[illegible]})}{(5^{-2})^{2}}$

   ☐ $\frac{(5^{[illegible]})\cdot 5}{5^{[illegible]}}$

4.5.6 Wrap-Up: Reflection

1. Use the emoji scale to indicate your level of confidence with evaluating numerical expressions with exponents.



2. Explain why you chose that level of confidence.